# Agricultural Crop Production and Yield Analytics Using FAOSTAT Data

## 1. Project Overview and Objective

### Project Overview

This project involves cleaning, transforming, and analysing agricultural crop production data using Python and Pandas. Data visualization techniques are used to identify crop production patterns across different areas and years. An interactive Power BI dashboard is developed to present key agricultural metrics, compare production values, and identify trends that support better agricultural planning and decision-making.

### Project Objective

The main objective of this project is to demonstrate data preprocessing, exploratory data analysis, and visualization techniques using Python, along with interactive dashboard development using Power BI. The project aims to identify production trends, compare agricultural performance across areas and crops, and present meaningful insights that support informed agricultural decision-making.

## 2.Data Source

- Source: FAOSTAT-Food and Agricultural Organization.
- Link: FAOSTAT
- Year/Timeline:2020-2025

## 3.PROBLEM STATEMENT

- To analyse crop production patterns across different areas and years to identify production trends and variations.
- To compare agricultural production across different crops and geographical areas.
- To identify changes in crop production over time and detect unusual values using statistical analysis and visualizations.
- To use data analytics and Power BI dashboards to present agricultural insights that support better agricultural planning and decision-making.

## 4. Attribute (Column /Features)

| Attribute Name | Data Type | Description |
|---|---|---|
| Domain Code | String | Unique code identifying the agricultural data domain |
| Domain | String | Name of the agricultural data domain |
| Area code(M49) | Integer/string | Code identifying the country |
| Area | String | Name of the country |

| | | |
|---|---|---|
| Element code | Integer | Code identifying the type of measurement |
| Element | Categorical | Type of agricultural measurement, such as production,<br>Yield. |
| Item code (CPC) | Integer/code | Code identifying the agricultural item or product. |
| Item | string | Name of the crop, livestock product, or agricultural item |
| Year | Integer | Year in which the agricultural data was recorded. |
| Unit | String | Code indicating the status or source of the data value |
| value | Numeric(decimal) | Recorded value of the agricultural measurement. |
| Flag | String | Code indicating the status or source of the data value. |

## 5.TOOLS & TECHNOLOGIES

- **python:** Data cleaning, transformation, and Pivot Tables.
- **Power BI:** Data modelling, DAX calculations, visualization, and interactive dashboard creation.

## 6. Data Pre-Processing (python / Power Query)

- Checked the dataset for missing values and duplicate records.
- Checked and corrected data types for the Year and Value columns.
- Standardized text fields such as Area, Item, and Element.
- Removed unnecessary columns only when they were not required for analysis.
- Checked extreme values in the Value column and retained valid observations.
- Filtered the Element column to select Production records for production-specific analysis.
- Fact & Dimension Tables: Converted the dataset into a Fact_ agricultural table
- created supporting dimension tables such as Dim_ Domain, Dim_ element, Dim _year, and Dim_ area for better analysis in Power BI.

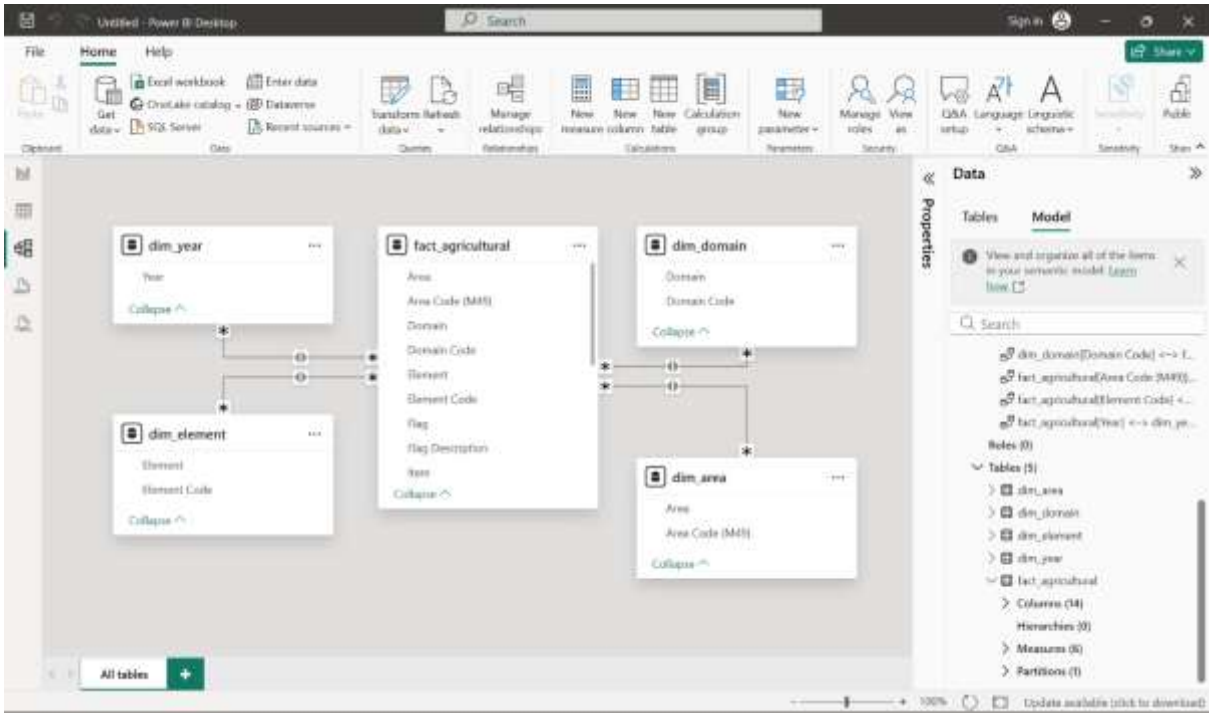
## 7. Data Modelling and DAX (Power BI)

- Created a Star Schema data model.
- Established relationships between the Fact_ agricultural table and dimension tables.
- Defined the correct cardinality (Many-to-One) for relationships.
- Used lookup tables for domain, year, area, and element information

**Calculated Columns & DAX Measures**

- **Total Production** – Calculates the total reported production value for records where the Element is Production.

- **Total Areas** – Counts the distinct areas represented in the dataset.
- **Average Production Value** – Calculates the average reported value for production records.



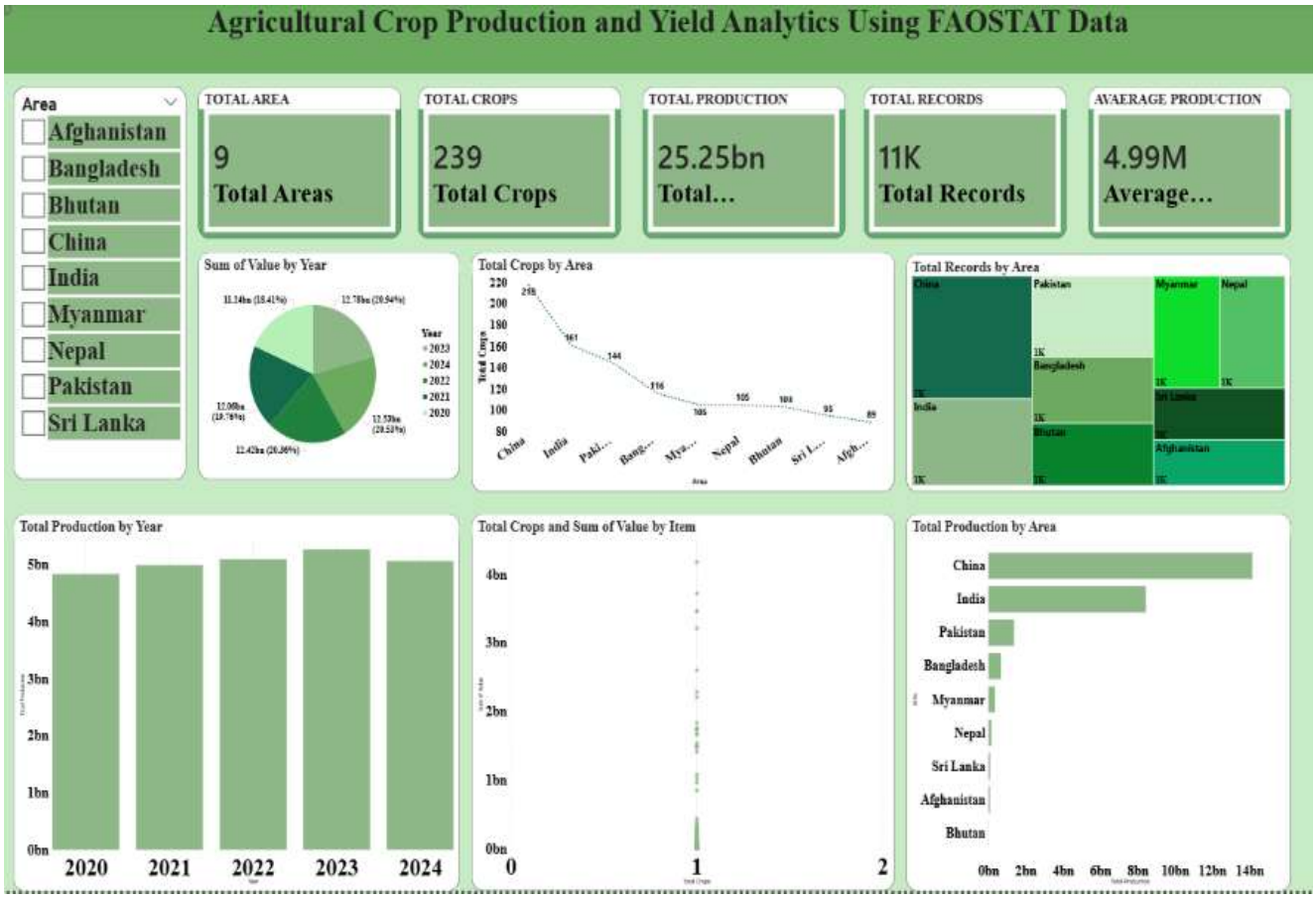

## 8. Analysis and Visualizations (Power BI)

- **KPI Cards:** Displays Total Area, Total Crops, Total Production, Total Records, and Average Production.
- **Area Slicer:** Filters agricultural data by country or area.
- **Pie Chart:** Shows the distribution of agricultural values by year.
- **Line Chart:** Compares the number of agricultural items across different areas.
- **Treemap:** Visualizes total records by area.
- **Column Chart:** Displays total production trends from 2020 to 2024.
- **Item Analysis Chart:** Compares agricultural items based on their production values.
- **Horizontal Bar Chart:** Ranks areas according to their reported production values.
- **Interactive Filters:** Allows users to explore data by selecting specific areas and years.
- **Data Labels and Titles:** Provides clear visual titles, labels, and values for better understanding.
- **Consolidated Dashboard:** Combines multiple visualizations into a single interactive Power BI report for agricultural data analysis.

## 9. Insights & Conclusions

### Key Findings

- Certain agricultural items recorded higher production values than others.
- Some areas, such as China and India, contributed higher summed production values than other areas in the dataset.
- Agricultural production values increased from 2020 to 2023, followed by a decrease in 2024.
- The dataset showed differences in production values across agricultural items and areas.
- Data quality, measurement units, and reporting coverage influenced the interpretation of production trends.




**Analysis Insights**

**Descriptive Analysis:** Summarized agricultural production values, total agricultural items, areas, and yearly trends using Power BI cards and charts.

**Diagnostic Analysis:** Compared agricultural items, areas, and years to investigate differences in reported production values and identify unusual patterns.

**Predictive Analysis:** Historical agricultural production data can be used to estimate future production trends using suitable forecasting models.

**Prescriptive Analysis:** Recommended investigating areas and agricultural items with declining production values, checking unusual records, and considering weather conditions, market demand, and resource availability to support agricultural planning.

## 10. Conclusion

The Agricultural Crop Production and Yield Analytics Dashboard project successfully demonstrated the use of python and Power Query for data cleaning and preprocessing and Power BI for data modelling, DAX calculations, and interactive visualizations. The dashboard provided

insights into agricultural production patterns, comparisons between areas and agricultural items, and yearly changes from 2020 to 2024. By transforming raw agricultural data into meaningful visual reports, the project supports better understanding of production trends and data-driven agricultural planning. Checking measurement units, data quality, and reporting coverage is essential for making reliable comparisons and informed decisions.